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HCAI 4301

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Assignment cleaning dataset

- What are the most significant data issues?
 - The most significant issues in the dataset were missing values in columns such as names, hire dates, and salaries. Also, there was inconsistent formatting in string fields such as state names, job titles, departments, and mixed date formats in the hire_date column. The different date styles made datetime conversion difficult. There were also duplicate records that needed to be corrected before step 3 of the analysis.
- What cleaning steps did you take?
 - I cleaned the dataset using .head(), .info(), and .describe() to identify data issues. I filled in missing salary values with the column mean, and any missing hire dates were filled in as well. Fields such as state, department, and job title were all standardized using string methods to ensure my formatting was consistent. I cleaned date values to reduce formatting inconsistencies present and converted them to datetime format. I normalized salaries using min-max scaling. I also removed duplicate records and merged the clean dataset and department codes table. Lastly, I created a pivot table to calculate the average salary per department.
- What would you do differently in a real-world scenario?
 - In a real-world scenario, I would avoid filling missing values without understanding the context of what I'm doing. I would confirm my understanding with higher-ups to ensure I am doing what they intended. I would also establish rules earlier to prevent inconsistent date formats and text entries submitted into the system. Finally, I would document all decisions made to clean the data.